
Optimization exam
November 2025

Exercise 1 (convex analysis). The objective of this exercise is to prove the strong concavity of the dual function of some optimization problems. Let $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be a convex function. We define the multifunction

$$\partial f(x) = \{s : f(y) \geq f(x) + s^T(y - x) \forall y \in X\}.$$

For a convex function $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ and $x, y \in X$, we have

$$f(y) = f(x) + \int_0^1 s(t)^T(y - x)dt$$

where $s(t)$ is an arbitrary selection in $\partial f(x(t))$ for any $x(\cdot)$ bijection from $[0, 1]$ to the segment $[x, y]$.

We recall that a function θ is strongly concave if and only if $-\theta$ is strongly convex and function f is strongly convex with constant of strong convexity $\alpha > 0$ with respect to norm $\|\cdot\|$ if and only if for all $0 \leq t \leq 1$, for all $x, y \in \text{dom}(f)$, we have

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) - \frac{\alpha t(1 - t)}{2} \|y - x\|^2.$$

In what follows, X is a nonempty convex subset of $\mathbb{R}^n$.

1) Show that function $f : X \rightarrow \mathbb{R}$ is strongly convex with constant of strong convexity $\alpha > 0$ with respect to norm $\|\cdot\|$ if and only if for every $x, y \in \text{dom}(f)$ we have

$$f(y) \geq f(x) + f'(x; y - x) + \frac{\alpha}{2} \|y - x\|^2,$$

where $f'(x; y - x)$ denotes the derivative of f at x in the direction $y - x$.

2) Show that if f is convex then $f'(x; d) = \max_{s \in \partial f(x)} s^T d$ and deduce that $f : X \rightarrow \mathbb{R}$ is strongly convex with constant of strong convexity $\alpha > 0$ with respect to norm $\|\cdot\|$ if and only if for every $x, y \in \text{dom}(f)$ we have

$$f(y) \geq f(x) + s^T(y - x) + \frac{\alpha}{2} \|y - x\|^2, \forall s \in \partial f(x).$$

3) Let $f : X \rightarrow \mathbb{R}$ be a convex function. Show that $f : X \rightarrow \mathbb{R}$ is strongly convex with constant of strong convexity $\alpha > 0$ with respect to norm $\|\cdot\|$ if and only if for every $x, y \in \text{dom}(f)$ we have

$$(s_y - s_x)^T(y - x) \geq \alpha \|y - x\|^2 \forall s_x \in \partial f(x), \forall s_y \in \partial f(y).$$

4) Show that if $f : X \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is strongly convex with constant of strong convexity $\alpha > 0$ with respect to norm $\|\cdot\|$ and $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear then $f + \ell$ is strongly convex on X with constant of strong convexity $\alpha > 0$ with respect to norm $\|\cdot\|$.

5) Let $X \subset \mathbb{R}^m, Y \subset \mathbb{R}^n$, be two nonempty convex sets. Let $\mathcal{A} : X \rightarrow Y$ be a linear operator and let $f : Y \rightarrow \mathbb{R}$ be a strongly convex function with constant of strong convexity $\alpha > 0$ with respect to a norm $\|\cdot\|_n$ on $\mathbb{R}^n$ induced by scalar product $\langle \cdot, \cdot \rangle_n$ on $\mathbb{R}^n$. Assume that $\text{Ker}(\mathcal{A}^* \circ \mathcal{A}) = \{0\}$. Show that $g = f \circ \mathcal{A}$ is strongly convex on X with constant of strong convexity $\alpha \lambda_{\min}(\mathcal{A}^* \circ \mathcal{A})$ with respect to norm $\|\cdot\|_m$.

6) Show that if f is differentiable and convex with

$$\|\nabla f(y) - \nabla f(x)\|_2 \leq M\|y - x\|_2 \quad \forall x, y \in \mathbb{R}^n \quad (1)$$

then for all $x, y \in \mathbb{R}^n$

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{M}{2}\|y - x\|_2^2. \quad (2)$$

7) We call a function that satisfies (2) a M -smooth function. Show that f is M -smooth if and only if $g(x) = \frac{M}{2}\|x\|_2^2 - f(x)$ is convex.

8) Assume that f is convex and satisfies (1). We want to show that it is co-coercive, meaning that

$$\langle y - x, \nabla f(y) - \nabla f(x) \rangle \geq \frac{1}{M}\|\nabla f(y) - \nabla f(x)\|_2^2. \quad (3)$$

a) Take $x_0 \in \mathbb{R}^n$ and define $\phi(y) = f(y) - \langle \nabla f(x_0), y \rangle$. Justify that x_0 is a global minimum of ϕ .

b) Justify that ϕ is M -smooth.

c) Applying relation (2) with f, y, x replaced by $\phi, x_0, y - \frac{1}{M}(\nabla f(y) - \nabla f(x_0))$ (justify that we can indeed apply the relation) deduce that (3) holds.

9) For $f : X \rightarrow \mathbb{R}$, we define its conjugate f^* by

$$f^*(s) = \sup_{y \in \mathbb{R}^n} s^T y - f(y).$$

a) Show that $f^*(s) = s^T x - f(x)$ if and only if $s \in \partial f(x)$.

b) Show that if $s \in \partial f(x)$ then $x \in \partial f^*(s)$.

c) If f is closed, i.e., if its level sets are closed, we will admit that $f^{**} = f$. Deduce that if f is closed then $s \in \partial f(x)$ if and only if $x \in \partial f^*(s)$.

10) We have:

Lemma 0.1 *If $\text{argmax}_{x \in \mathbb{R}^n} \{s^T x - f(x)\}$ is a singleton then f^* is differentiable at s and $\partial f^*(s) = \{\nabla f^*(s)\} = \text{argmax}_{x \in \mathbb{R}^n} \{s^T x - f(x)\}$.*

Using Lemma 0.1 and 9) show that if f is closed and strongly convex with constant of strong convexity $1/M$ with respect to $\|\cdot\|_2$ then f^* satisfies (1).

11) Using 8)-9) show that if f is convex and satisfies (1) then f^* is strongly convex with constant of strong convexity $1/M$ with respect to $\|\cdot\|_2$.

12) Consider the optimization problem

$$\begin{cases} \inf f(x) \\ Ax \leq b \end{cases} \quad (4)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $b \in \mathbb{R}^q$, and A is a $q \times n$ real matrix.

Let θ be the dual function of (4) given by

$$\theta(\lambda) = \inf_{x \in \mathbb{R}^n} \{f(x) + \lambda^T(Ax - b)\}, \quad (5)$$

for $\lambda \in \mathbb{R}^q$. Assume that the rows of matrix A are independent, that f is convex, differentiable, and ∇f is Lipschitz continuous with constant $L \geq 0$ with respect to norm $\|\cdot\|_2$.

Using the previous results, show that dual function θ is strongly concave on $\mathbb{R}^q$ with constant of strong concavity $\frac{\lambda_{\min}(AA^T)}{L}$ with respect to norm $\|\cdot\|_2$ on $\mathbb{R}^q$.

Exercise 2 (subgradient algorithm). For a set $X \subset \mathbb{R}^n$ and $\bar{x} \in X$, the normal cone $\mathcal{N}_X(\bar{x})$ to X at $\bar{x}$ is given by

$$\mathcal{N}_X(\bar{x}) = \{z \in \mathbb{R}^n : (x - \bar{x})^T z \leq 0 \forall x \in X\}.$$

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. To solve

$$f_* = \min_{x \in \mathbb{R}^n} f(x) \quad (6)$$

the subgradient method generates the sequence (x_k) given by

$$x_{k+1} = x_k - \alpha_k g_k, \quad k \geq 1,$$

where g_k is an arbitrary vector in $\partial f(x_k)$ and $\alpha_k > 0$. We will denote

$$f_k = \min_{1 \leq i \leq k} f(x_i).$$

We assume:

(H1) f_* is finite and attained.

(H2) There is a constant M such that for every x and $g \in \partial f(x)$ we have $\|g\| \leq M$.

Let x_* be an optimal solution of (6) and let R such that $\|x_1 - x_*\|_2 \leq R$.

1) Using the definition of $\partial f(x_k)$, show that

$$\|x_{k+1} - x_*\|_2^2 \leq \|x_k - x_*\|_2^2 - 2\alpha_k(f(x_k) - f_*) + \alpha_k^2 \|g_k\|_2^2. \quad (7)$$

2) Deduce from (7) that

$$\|x_{k+1} - x_*\|_2^2 \leq R^2 - 2 \sum_{i=1}^k \alpha_i (f(x_i) - f_*) + M^2 \sum_{i=1}^k \alpha_i^2. \quad (8)$$

3) Deduce from (8) that

$$f_* \leq f_k \leq f_* + \frac{R^2 + M^2 \sum_{i=1}^k \alpha_i^2}{2 \sum_{i=1}^k \alpha_i}. \quad (9)$$

4) Let U be the function given by

$$U(\alpha_1, \dots, \alpha_k) = \frac{R^2 + M^2 \sum_{i=1}^k \alpha_i^2}{2 \sum_{i=1}^k \alpha_i}.$$

Show that U is convex and find a minimum of U on $\mathbb{R}^k$.

5) Deduce from 3)4) the sequence of step sizes (α_k) providing the best upper bound for $f_k - f_*$ in (9). For the choice of this sequence, show that at least $\frac{R^2 M^2}{\varepsilon^2}$ iterations are necessary to obtain an ε -optimal solution of (6) using the subgradient method.

6) Application. We want to find a point in a given intersection of closed convex sets

$$C = C_1 \cap \dots \cap C_m.$$

a) Justify that this problem can be modelled as a problem of form (6) with $f(x) = \max(\text{dist}(x, C_1), \dots, \text{dist}(x, C_m))$ where $\text{dist}(x, A) = \inf_{y \in A} \|y - x\|_2$.

b) Fix $x_0 \in \mathbb{R}^n$ and let P_j be the projection operator on C_j . Let j such that $f(x_0) = \text{dist}(x_0, C_j)$. Justify that $0 \in \partial f(x_0)$ if $x_0 \in C_j$ and

$$\frac{x_0 - P_j(x_0)}{\|x_0 - P_j(x_0)\|_2} \in \partial f(x_0)$$

otherwise. Write the subgradient method to minimize f on $\mathbb{R}^n$.